

Machine Learning and Deep Learning Capstone Project Report

1. Introduction

Sentiment analysis is a Natural Language Processing (NLP) task that aims to identify the emotional meaning of text. In this project, a machine learning and deep learning approach was applied to classify movie reviews as positive or negative.

The project uses the IMDB Movie Reviews dataset, where each review is labeled with a sentiment. The goal is to build classification models that can automatically predict whether a review expresses a positive or negative opinion.

2. Dataset Description

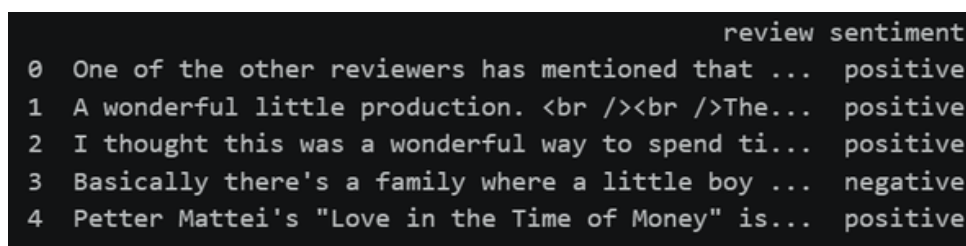
The dataset used in this project is the IMDB Dataset, which contains movie reviews and their corresponding sentiment labels.

Each record contains:

- Review: The text written by the user.
- Sentiment: The classification label:
 1. Positive
 2. Negative

The dataset was prepared by removing missing values and processing the text before training the models.

Figure 1: Dataset Preview



	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

3. Data Preprocessing

Before training the models, several preprocessing steps were performed:

1. Text Cleaning:
2. Converted all text to lowercase.
3. Removed unnecessary characters and symbols.
4. Removed stop words to reduce noise.

5. Text Transformation because machine learning models cannot directly understand text, and the reviews were converted into numerical features using TF-IDF (Term Frequency-Inverse Document Frequency).

TF-IDF assigns importance scores to words based on their frequency in the review and the dataset.

Data Splitting: The dataset was divided into:

- 80% Training data
- 20% Testing data

The training data was used to build the models, while the testing data was used to evaluate performance.

4. Machine Learning Models

Three machine learning classification models were implemented:

4.1 Logistic Regression

Logistic Regression was used as a baseline classification model. It predicts whether a review is positive or negative based on the TF-IDF features.

4.2 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

4.3 K-Nearest Neighbors (KNN)

KNN classifies reviews by comparing them with similar examples from the training dataset.

5. Model Evaluation

The models were evaluated using several performance metrics:

Accuracy

Measures the percentage of correct predictions made by the model.

Precision

Measures how many predicted positive results were actually positive.

Recall

Measures how many actual positive reviews were correctly identified.

F1 Score

Combines precision and recall into one evaluation metric.

Confusion matrices were also generated to analyze the number of correct and incorrect predictions.

6. Deep Learning Model

A Neural Network model was created using TensorFlow and Keras.

The architecture includes:

- Input layer based on TF-IDF features.
- Dense layer with 128 neurons using ReLU activation.
- Dropout layer to reduce overfitting.
- Dense layer with 64 neurons.
- Output layer with sigmoid activation for binary classification.

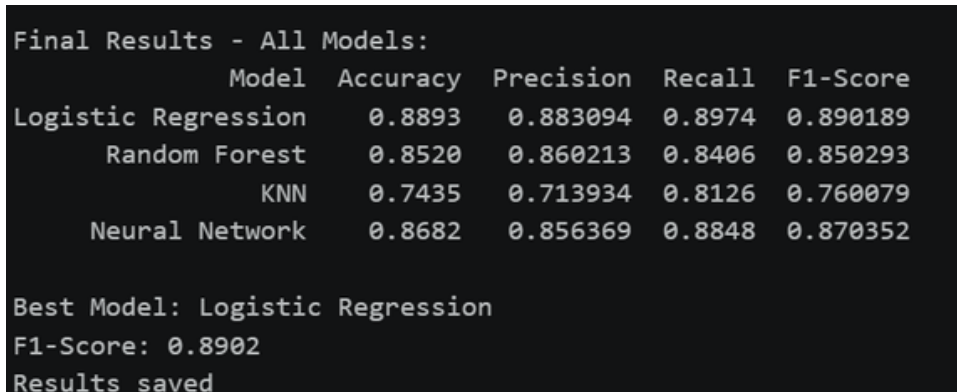
The model was trained for multiple epochs and evaluated using the test dataset.

7. Results and Comparison

The models were evaluated using accuracy, precision, recall, and F1 score.

The best model was selected based on the highest evaluation metrics.

The final results are shown below:



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Final Results - All Models:
      Model  Accuracy  Precision  Recall  F1-Score
Logistic Regression  0.8893   0.883094   0.8974   0.890189
      Random Forest  0.8520   0.860213   0.8406   0.850293
              KNN    0.7435   0.713934   0.8126   0.760079
      Neural Network  0.8682   0.856369   0.8848   0.870352

Best Model: Logistic Regression
F1-Score: 0.8902
Results saved
```

8. Conclusion

In this project, Natural Language Processing techniques were combined with machine learning and deep learning algorithms to perform sentiment analysis on movie reviews.

The results demonstrate that text preprocessing and TF-IDF feature extraction are important steps for converting human language into a format that machine learning models can understand.

The implemented models successfully classified movie reviews into positive and negative categories, showing the effectiveness of machine learning methods in text classification tasks.

9. What I Learned

Through this project, I learned:

- How to preprocess and clean text data.
- How NLP techniques convert text into numerical features.
- How to train and evaluate classification models.
- How to use confusion matrices and evaluation metrics.
- How to build a neural network using TensorFlow and Keras.
- The differences between traditional machine learning and deep learning approaches.

10. Project Code

This GitHub repository contains the complete project details, including all code, preprocessing steps, model training, evaluation, and implementation.

<https://github.com/abdurahmanmalyasin-wq/Machine-Learning-and-Deep-Learning-Capstone-Project.git>